

# Marginal abatement cost curves for policy recommendation – A method for energy system analysis

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## ABSTRACT

The transport sector is seen as one of the key factors for driving future energy consumption and greenhouse gas (GHG) emissions. In order to rank possible measures marginal abatement cost curves have become a tool to graphically represent the relationship between abatement costs and emission reduction. This paper demonstrates how to derive marginal abatement cost curves for well-to-wheel GHG emissions of the transport sector considering the full energy provision chain and the interlinkages and interdependencies within the energy system. Presented marginal abatement cost curves visualize substitution effects between measures for different marginal mitigation costs. The analysis makes use of an application of the energy system model generator TIMES for South Africa (TIMES-GEECO). For the example of Gauteng province, this study exemplary shows that the transport sector is not the first sector to address for cost-efficient reduction of GHG emissions. However, the analysis also demonstrates that several options are available to mitigate transport related GHG emissions at comparable low marginal abatement costs. This methodology can be transferred to other economic sectors as well as to other regions in the world to derive cost-efficient GHG reduction strategies

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## 1. Introduction and background

In the Kyoto protocol the participating countries committed themselves to reduce greenhouse gas (GHG) emissions by 5.2% below the 1990 level by 2012. According to the IPCC Fifth Assessment Report (IPCC, 2014) the transport sector was responsible for about 14% (6.9 Gt CO<sub>2</sub>e) of global GHG emissions in 2010, which is an increase of almost 50% since 1990. In the same timeframe the global oil demand increased by about 30% (IEA, 2012). This trend is likely to continue where the transport sector (especially in the emerging economies) is the main driver for increased GHG emissions (IEA, 2012).

The global reduction of GHG emissions can probably only be achieved if the economic burden is minimized. Marginal abatement cost (MAC) curves as graphical representations of the relationship between abatement costs and emission level have become a tool to determine the appropriate set of measures to reach the desired carbon reduction target. MAC curves have so far been applied to numerous demand sectors and regions in the world (see e.g., Kesicki, 2013, 2012a,b; McKinsey&Company, 2009; Remme, 2006; Schrotten et al., 2012; Bockel et al., 2012; MacLeod et al., 2010).

In principle there are three levers for mitigating transport related GHG emissions: the first option is reducing demand for motorized transport (kilometers traveled per mode), e.g., by

changing the modal split through enhancing public transport or freight mass transportation services (see e.g., Fiorio et al., 2013; Redman et al., 2013; Galilea and Medda, 2010; Buehler and Pucher, 2011; Albalade and Bel, 2010; John and Kurth, 1995). The second possibility lies in the reduction of vehicle emission intensity (i.e., tank-to-wheel (TTW) emissions) which can be achieved through increased vehicle efficiency (such as in hybrid electric vehicles) or by using alternative fuels (e.g., by using low-carbon fuels or by balancing carbon credits for biomass production). Finally, there is the possibility of decreasing the emission intensity of fuel provision (i.e., well-to-tank (WTT) emissions), e.g., through alternative fuel provision measures such as biofuels, natural gas or carbon capture and storage. Many studies are available analyzing either one or many of these aspects in more detail (see e.g., Bruchof, 2013; Özdemir, 2012; Flachsland et al., 2011; IEA, 2010; Gül, 2008).

This paper will demonstrate how to derive marginal abatement cost curves for well-to-wheel GHG emissions of the transport sector to demonstrate which measures should initially be applied and therewith to be supported through legislation. The analysis makes use of the energy system model generator, TIMES, and considers the full energy provision chain and the interlinkages and interdependencies within the energy system. To make the presented MAC curves more transparent this paper will also show the changes and substitution effects in marginal abatement cost curves by attributing sectoral effects in energy provision to energy use in the demand sector.

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**Symbols and indices**

|                         |                                                                                                                           |                    |                                                                                   |
|-------------------------|---------------------------------------------------------------------------------------------------------------------------|--------------------|-----------------------------------------------------------------------------------|
| $c$                     | commodity                                                                                                                 | $n$                | transport powertrain                                                              |
| $c_i$                   | commodity of type $i$ . Fuel products                                                                                     | $N_2O$             | nitrous oxide                                                                     |
| $c_{i^*}$               | commodity of type $i^*$ . Feedstock for conversion process $p_i^{CONV}$                                                   | $out$              | output                                                                            |
| $c \in pcg(p)$          | commodity which is element of the primary commodity group of process $p$                                                  | $p$                | process                                                                           |
| $C_i$                   | cost of technology $i$                                                                                                    | $p \in p_i$        | group of processes $p_i$ of a similar type                                        |
| $C_{Ref}$               | cost of reference technology                                                                                              | $p_i^{CONV}$       | conversion process for fuel provision of type $i$                                 |
| $CH_4$                  | methane                                                                                                                   | $p_i^{TRA}$        | transport processes by mode $m$                                                   |
| $CO_2$                  | carbon dioxide                                                                                                            | $p_{m,n}^{TRA}$    | transport processes by powertrain $n$                                             |
| $CONV$                  | energy conversion sector                                                                                                  | $p_{m,n}^{TRA}$    | transport process of mode $m$ with powertrain technology $n$                      |
| $DIST$                  | energy delivery and distribution                                                                                          | $pcg$              | primary commodity group                                                           |
| $E_i$                   | emissions of technology $i$                                                                                               | $PRIM$             | primary energy provision                                                          |
| $E_{Ref}$               | emissions of reference technology                                                                                         | $r$                | region                                                                            |
| $FLO(r, v, t, p, c, s)$ | flow of commodity $c$ of process $p$ with vintage year $v$ in current period $t$ and time slice segment $s$ in region $r$ | $s$                | time slice segment                                                                |
| $g$                     | type of primary energy provision process                                                                                  | $t$                | (current) time period                                                             |
| $GHG$                   | greenhouse gases: $CO_2$ , $CH_4$ and $N_2O$                                                                              | $TEC$              | transport related energy consumption                                              |
| $i$                     | fuel product (final energy carrier)                                                                                       | $TEP$              | transport related energy provision                                                |
| $i^*$                   | feedstock for conversion process (primary energy carrier)                                                                 | $TRA$              | transport sector                                                                  |
| $in$                    | input                                                                                                                     | $TTW$              | tank-to-wheel emissions                                                           |
| $k$                     | marginal abatement cost [EUR <sub>10</sub> /t CO <sub>2</sub> e]                                                          | $v$                | construction time period (vintage)                                                |
| $l$                     | type of conversion process                                                                                                | $WTT$              | well-to-tank emissions                                                            |
| $m$                     | transport mode                                                                                                            | $WTW$              | well-to-wheel emissions                                                           |
|                         |                                                                                                                           | $wtt$              | well-to-tank emission factor                                                      |
|                         |                                                                                                                           | $z$                | objective function                                                                |
|                         |                                                                                                                           | $\gamma$           | transport share                                                                   |
|                         |                                                                                                                           | $\eta(p_i^{DIST})$ | efficiency of distribution process for commodity $c_i$ , i.e. distribution losses |

The analysis will be carried out for Gauteng Province, South Africa as an example. The South African government has committed to reduce greenhouse gas emissions by 34% till 2020 and by 42% till 2025 against a business as usual trajectory. A carbon tax based on CO<sub>2</sub> equivalents for all energy consuming sectors has been proposed in order to achieve this ambitious goal. However, the rate at which such a tax should be applied is still under discussion as well as possible exemptions for energy intensive industries (DNT, 2013).

The metropolitan region of Gauteng is the economic hub of South Africa with a third of the national GDP and a fifth of the population. The transport sector is responsible for about 25% of total GHG emissions, taking into account emissions from the energy supply (e.g., petroleum refining and synthetic fuels) (Tomaschek et al., 2012b). Gauteng Province sees an opportunity to be a forerunner for the South African climate protection strategy but has not yet developed a clear strategy how the transport sector could and should develop in future (DLGH, 2010).

The paper is structured as follows: Section 2.1 presents recent studies covering MAC curves. Furthermore, technology oriented bottom-up approaches using energy system models are described, where in both cases focus is given on those studies analyzing the transport sector or transport related issues in more detail. Section 2.2 presents the basic framework of the TIMES-GEECO model and its structure as well as technology detail in the transport sector and in the fuel provision sector. Section 2.3 shows how MAC curves for the transport sector can be derived which incorporate fuel provision, interdependencies and interlinkages between options, transport related ancillary effects within the energy system as well as the substitution effects between options. The definition of the scenarios analyzed can be found in chapter 3. Chapter 4 presents the results of the model application and shows how to cost-optimally mitigate transport related well-to-wheel GHG emissions in Gauteng and in South Africa. Chapter 5 concludes and gives detailed policy recommendations.

## 2. Methodology

### 2.1. Literature review

Two fundamental approaches for deriving MAC curves can be distinguished: firstly, the calculation of marginal abatement costs of alternative measures against a reference and subsequent ranking (static approach) and, secondly, the application of mathematical models of the economy or energy system, i.e., computable general equilibrium (GCE) models or bottom-up approaches using energy models.

Static approaches calculate the GHG emissions ( $E$ ) before and after the implementation of an alternative technology ( $i$ ) and associate the emission abatement to the costs of the measure in comparison to a reference technology ( $Ref$ ) (Eq. (1)).

$$MAC(i) = \frac{C_i - C_{Ref}}{E_{Ref} - E_i} \quad (1)$$

One of the most well-known presentations of such a MAC curves for the transport sector is probably the work of McKinsey&Company (2009) where they present abatement cost curves for different regions in the world and also a global MAC curve for the transport sector. The work of Schrotten et al. (2012) laid its emphasis on the freight sector, and derived detailed MAC curves for heavy duty vehicles covering different delivery scenarios, distinguishing alternative powertrains, engine modifications as well as enhancements of vehicle aerodynamics and rolling resistance. An example for a sector wide MAC curve can be found in Wächter (2013) who identified some no-cost mitigation measures available for the transport sector of Austria but did not include emissions associated with energy provision. Telsnig et al. (2013) analyzed a broad set of abatement measures for the South African energy system based on a static approach. Static approaches are relatively easy to realize. In principle the analysis requires only the cost of two compared

measures as well as their emissions level. However, static approaches fail to integrate the interdependencies and interlinkages between measures and bare the possibility of inconsistencies.

Energy system models, on the contrary, include those interlinkages on an intertemporal basis and, thus, provide a consistent baseline. By using energy system models it is possible to deduce MAC curves by applying carbon (or GHG) taxes on all energy provision and energy consuming sectors. In this case the marginal abatement costs of different measures can be derived by performing model runs with taxes (*tax*) allocated to carbon (or GHG) emissions and checking which technologies are part of the model solution given a certain tax (Eq. (2)). The difficulty in this approach can be seen in calculating the corresponding emission mitigation allocated to the implementation of measure, *i*, as the whole system is altered simultaneously.

$$E(i) = f(\text{MAC} = \text{tax}) \quad (2)$$

Energy system models are largely applied to identify cost optimal measures for GHG mitigation. Frequently used energy system models are for example MARKAL/TIMES, MESSAGE, PRIMES and the LEAP model. Energy system models can be differentiated into bottom-up and top-down models. While the latter generally apply a broad and aggregated approach to analyze interlinkages between the energy sector and the remainder of the economy, the former use a very detailed technology specific approach to derive conclusions.

Nakata et al. (2011) gave a good overview about the general energy system modeling approach. Furthermore, they presented several technology options for GHG mitigation including the transport sector. Recent publications about energy system models covering the transport sector assessed certain aspects of transport related GHG abatement options without using the concept of marginal abatement cost curves such as hydrogen supply and use (see e.g., Hedenus et al., 2010; Contreras et al., 2009), alternative fuels for personal transport (see e.g., Gül et al., 2009; Bruchof, 2013; Özdemir, 2012) and /or electric mobility (see e.g., Densing et al., 2012). The International Energy Agency summarized more than 350 publications focusing on applications of the TIMES-MARKAL model generator for the years 2008–2010 (IEA, 2011). The focus of those publications clearly lies on the European or US American context and only about 6% of the publications investigated a regional or local context. Furthermore, little attention has been given to transport orientated analysis, which clearly underlines the need for research in that area.

Remme (2006) presented GHG mitigation pathways for the German energy system. The analysis was carried out using a TIMES model for Germany with high detail in energy provision technologies, comparing different scenarios and price assumptions. Simões et al. (2008) presented an application of the TIMES model generator for Portugal to derive CO<sub>2</sub> abatement cost curves for electricity supply, showing inter alia the effects of a nuclear phase out. In terms of abatement options for the transport sector the model considers some basic options such as biofuels and gas. Chen (2005) presented MAC curves derived from a MARKAL-MACRO model for China considering the effects of taking carbon mitigation actions on the national GDP. He points out the importance for developing countries to maintain low per capita carbon intensity rather than restricting themselves to strict emission limits. Some excellent examples of using energy system models for deriving MAC curves have been provided by Kesicki for the transport sector of the UK (see e.g., Kesicki, 2012a,b, 2010). In his work he presents how an energy system model can cover the interlinkages, interdependencies and ancillary effects of transport related mitigation options using a UK MARKAL model.

A principle issue in all these MAC curves is that substitution effects between measures are not shown. For example CCS at lower marginal abatement cost might drop out for a zero carbon option at

higher marginal cost. Moreover, the allocation of effects and measures to one economic sector is not always clear: For example when analyzing the transport sector, changes in the energy provision are obviously partly attributable to transport activity (e.g., change in emissions for fuel provision). Though, feedback loops like ancillary energy needs (e.g., electricity) make allocation to one sector difficult. Decomposition analysis is a possibility to overcome some of the shortages of energy system model based MAC curves and attribute emission mitigation to certain effects such as fuel switch, fuel intensity, technology change, energy efficiency or activity level (Kesicki, 2012a,b; Murphy and Jaccard, 2011; Diakoulaki et al., 2006). Such decomposition approaches however do not enable to allocate technology specific measures in energy provision to their sector specific service provision such as in the transport sector.

It can be summarized that MAC curves are a widely applied methodology in order to provide a basis for policy recommendations on how to mitigate carbon emissions. However, different approaches to derive such abatement costs are available which differ in their complexity and ability to reproduce real-world interrelationships. Furthermore, transparency in the depth of technical detail or full consideration of greenhouse gases not always receives necessary attention. Finally, most of the MAC curves in literature usually only show carbon emission and not GHG emissions and do therewith exclude a certain share of climate relevant gases.

## 2.2. The TIMES model generator and its application for Gauteng and South Africa

The TIMES model generator is a technology based partial equilibrium model that assumes perfect foresight and perfect competition (Loulou et al., 2005). In TIMES the objective (*z*) is to minimize the total system costs (see Eq. (3)). Cost components of the objective function include inter alia capital costs for investing into or dismantling processes; fixed operating and maintenance costs and other annual costs; variable costs dependent on the activity (production level) of a process as well as those dependent on flow costs of commodities (e.g., CO<sub>2</sub> transport, delivery costs); import expenditures and export revenues; taxes and subsidies associated with commodity flows, process activities or investments as well as the salvage value of processes at the end of the planning horizon. All costs are discounted to the beginning of the base year.

$$z: C_0 = \sum_{a=1}^b C_j \cdot x_j + C \stackrel{!}{=} \min \quad (a = 1, \dots, b) \quad (3)$$

The TIMES-GEECO model, which is an abbreviation for “TIMES-Gauteng Energy and Emission Cost Optimization” model, is an application of the TIMES model generator. This model, which has been developed at the IER, University of Stuttgart, encompasses the entire energy system of Gauteng and the energy supply structure of South Africa (refer e.g., to Haasz et al., 2013; Tomaschek et al., 2012a; Dobbins et al., 2009 for further information on this model).

TIMES-GEECO incorporates numerous investment options for all the relevant demand sectors (i.e. industry, transport, commerce, residential and public buildings), as well as for the energy supply, with a high level of detail in electricity and fuel provision. Features which characterize this model application are inter alia the consideration of building structures and income groups in the residential sector, the possibility for expanding public transport infrastructure to alter the modal shift as well as the high level of detail in alternative production routes in the mining and steel industry. Furthermore, the model allows for carbon capture and storage (CCS). TIMES-GEECO integrates 42 timeslices that include three seasons (summer, winter and intermediate), two typical days for each season (weekday and weekend) and seven segments that aggregate hours. The time-horizon of the model is the year 2040.

In the TIMES-GEECO model a broad set of possible vehicle powertrain technologies has been incorporated. The reference technologies are conventional internal combustion engines (ICE) which are used in most motorized vehicles today in the world. However, measures to improve ICEs such as downsizing and turbocharging (allowing high torque and load by reduced engine capacity), have been considered for future evolution of this technology. Furthermore, alternative concepts are part of TIMES-GEECO such as electric vehicles (differentiated by the degree of electrification and by means of energy storage) and dedicated biofuel and flex-fuel vehicles (e.g., E85 and B100), see Table 1. The analysis has been carried out for various mobility modes of which some are characteristic for the study area of Gauteng such as a bus rapid transit system (BRT), a high-speed rail system (i.e. the Gautrain) and minibuses (Table 2). The fuel sector of TIMES-GEECO incorporates various types of liquid and gaseous fuels with both renewable and fossil sources (Table 3).

Fig. 1 illustrates how the different sectors are interlinked in the TIMES-GEECO model. PRIM refers to “primary energy provision”; CONV to “energy conversion”; DIST to “energy delivery and distribution” and TRA refers to the transport sector.

### 2.3. Marginal abatement cost curves for transport WTW GHG emissions

In order to derive the marginal abatement cost curves taxes are applied on greenhouse gas emissions. The carbon equivalent of those emissions ( $c_{GHG}$ ) is calculated by considering the three most important greenhouse gases:  $CO_2$ ,  $CH_4$  and  $N_2O$ . The global warming potential was calculated using figures from IPCC (2001) (see Eq. (4)) in line with available emission inventories for South Africa (DEA, 2013) and for Gauteng (Tomaschek et al., 2012b) which are based on the same guidelines.

$$c_{GHG} [g CO_2e] = c_{CO_2} + 23 \cdot c_{CH_4} + 296 \cdot c_{N_2O} \quad (4)$$

To quantify the emission mitigation of the applied measures

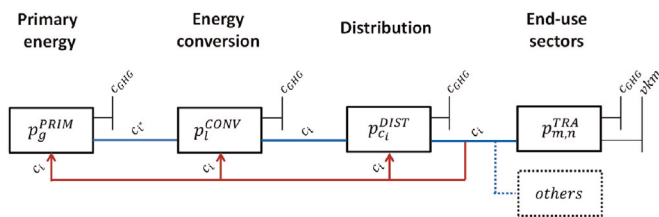


Fig. 1. Energy process chain in the TIMES-GEECO model (schematic).

Table 1  
Vehicle technologies for road transport considered in the TIMES-GEECO model.

| Powertrain technology<br>$p_n^{TRA}$ | Description                                                                                                                                                     | Fuel used $c_i$                                        |
|--------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------|
| ICE                                  | Internal combustion engines are the main engine form of current vehicles in the world. ICEs can be modified for gas use or dedicated biofuel usage at low cost. | Petrol, diesel, biodiesel, ethanol, CNG, LPG, hydrogen |
| Mild hybrid                          | In a mild-hybrid vehicle, the conventional engine is supported by an electric motor during peak load. Mild hybrid vehicles are not capable of electric driving. | Petrol, diesel                                         |
| Full hybrid                          | Fully hybrid vehicles are capable of electric driving over short distances (about 2–10 km). Thus battery and electric motor have to be sized bigger.            | Petrol, diesel                                         |
| PHEV                                 | Plug-in hybrid electric vehicles are capable of electric driving over long distances (about 20–50 km) and can be recharged from the grid.                       | Petrol, diesel, electricity                            |
| BEV                                  | Battery electric vehicles are pure electric vehicles, which use batteries as energy storage. BEVs cause no local emissions.                                     | Electricity                                            |
| FCHEV                                | In difference to BEVs fuel-cell hybrid electric vehicles use hydrogen as energy storage, which is converted to electricity in a fuel-cell.                      | Hydrogen                                               |

Table 2  
Transport modes considered in TIMES-GEECO.

| Category                       | Mode of transport $p_m^{TRA}$                                                             |
|--------------------------------|-------------------------------------------------------------------------------------------|
| Non-motorized transport        | Walking<br>Bicycles                                                                       |
| Individual passenger transport | Motorcycles<br>Passenger Cars<br>SUVs                                                     |
| Public transport               | Minibus Taxi<br>Buses & BRT<br>Passenger rail & Gautrain (high-speed railway)<br>Aviation |
| Freight transport              | Light duty vehicles (LDV)<br>Heavy duty vehicles (HDV)<br>Freight rail<br>Aviation        |

the total (i.e., well-to-wheel, WTW) GHG emissions of the transport sector are taken into account (Eq. (5)). These consist of GHG emissions directly related to fuel combustion in vehicles (i.e., tank-to-wheel, TTW) as well as of GHG emissions that are a result of indirect emissions (i.e., well-to-tank, WTT) that go in hand with fuel and energy provision for transport activities.

$$\underbrace{WTW_{GHG}^{TRA}}_{\text{Total transport related GHG emissions}} [g CO_2e] = \underbrace{TTW_{GHG}^{TRA}}_{\text{Direct emissions related to fuel combustion}} + \underbrace{WTT_{GHG}^{TRA}}_{\text{Indirect emissions related to fuel and energy provision}} \quad (5)$$

Tank-to-wheel GHG emissions of the transport sector ( $TTW_{GHG}^{TRA}$ ) can easily be quantified as they are output flows ( $FLO_{out}$ ) of the modeled transport processes  $p_{m,n}^{TRA}$  (Eq. (6)).

$$\underbrace{TTW_{GHG,k}^{TRA}}_{\text{Total transport TTW emissions}} [g CO_2e] = \underbrace{\sum_v \sum_s FLO_{out}(r, v, t, p_{m,n}^{TRA}, c_{GHG}, s)_k}_{\text{Sum of GHG emission flows for all transport processes}} \quad (6)$$

where  $r$  = region;  $t$  = period (here: 2040);  $p_{m,n}^{TRA}$  = transport process of mode  $m$  (Table 2) with powertrain technology  $n$  (Table 1),  $c_{GHG}$  = greenhouse gas emissions. The TIMES parameters  $s$  (time-slice) and  $v$  (vintage) are not relevant for this analysis and are therefore summed up. The index  $k$  reflects to the GHG taxes applied,  $k=\{0, \dots, 250\}$  [EUR10/t CO<sub>2</sub>e]. TRA indicates the transport sector.

To allocate well-to-tank GHG emissions to the transport sector emission factors are calculated to be applied to the energy consumption in the transport sector. Emission factors for energy conversion (CONV) and energy distribution (DIST) can be aggregated for similar energy carriers ( $c_i$ ) (Eq. (7)). Furthermore, primary energy provision (e.g., coal mining) can cause additional emissions which are to a certain share attributable to transport sector activity. The GHG emission factor for a primary energy provision process can be expressed as shown in Eq. (8)

$$\underbrace{wtt_{c_i}^{CONV,DIST}}_{\text{Specific WTT emissions of a final energy carrier}} [g \text{ CO}_2e/J] = \frac{\sum_p \sum_r \sum_v \sum_s FLO_{out}(r, v, t, p, c_{GHG}, s)_k}{\sum_p \sum_{c_i} \sum_r \sum_v \sum_s FLO_{out}(r, v, t, p, c_i, s)_k} \quad (7)$$

Quotient of GHG emission flows of and total provision of a final energy carrier

$; p \in \{p_l^{CONV}, p_{c_i}^{DIST}\} \wedge c_i \in pcg(p)$

$$\underbrace{wtt_{c_i^*}^{PRIM}}_{\text{Specific WTT emissions of a primary energy carrier}} [g \text{ CO}_2e/J] = \frac{\sum_p \sum_r \sum_v \sum_s FLO_{out}(r, v, t, p, c_{GHG}, s)_k}{\sum_p \sum_{c_i^*} \sum_r \sum_v \sum_s FLO_{out}(r, v, t, p, c_i^*, s)_k} \quad (8)$$

Quotient of GHG emission flows of and total provision of a primary energy carrier

$; p \in p_g^{PRIM} \wedge c_i^* \in pcg(p)$

With  $p_l^{CONV}$  = conversion process of type  $l$ ;  $p_g^{PRIM}$  = primary energy provision process of type  $g$ ;  $p_{c_i}^{DIST}$  = distribution process for commodities  $c_i$  and  $c_i^*$  = commodity of type  $i$ ;  $c_i^*$  = commodity of type  $i^*$

1. The transport related energy provision ( $TEP^{TRA}$ ) of all final energy carriers  $c_i$ , Eq. (9)

$$\underbrace{TEP_{c_i}^{TRA}}_{\text{Total transport related provision of a final energy carrier}} [J] = \frac{1}{\underbrace{\eta(p_{c_i}^{DIST})}_{\text{Distribution losses}}} \cdot \underbrace{\sum_{c_i} \sum_{p_{m,n}^{TRA}} \sum_v \sum_s FLO_{in}(r, v, t, p_{m,n}^{TRA}, c_i, s)_k}_{\text{Transport energy consumption of a final energy carrier}} ; c_i \in pcg(p_l^{CONV}) \quad (9)$$

2. The transport related energy provision of all primary energy carriers  $c_i^*$ , Eq. (10)

$$\underbrace{TEP_{c_i^*}^{TRA}}_{\text{Total transport related provision of a primary energy carrier}} [J] = \underbrace{\gamma_k}_{\text{Share of the transport sector}} \cdot \underbrace{\sum_{c_i^*} \sum_{p_l^{CONV}} \sum_v \sum_s FLO_{in}(r, v, t, p_l^{CONV}, c_i^*, s)_k}_{\text{Total consumption of a primary energy carrier for final energy provision}} ; c_i^* \in pcg(p_g^{PRIM}) \quad (10)$$

3. The transport related consumption of all final energy carriers  $c_i$  used as ancillary energy for primary energy provision, energy conversion and energy distribution, Eq. (11)

$$\underbrace{TEC_{c_i}^{PRIM,CONV,DIST}}_{\text{Total transport related ancillary energy consumption}} [J] = \left[ \underbrace{\sum_{p_g^{PRIM}} \sum_v \sum_s FLO_{in}(r, v, t, p_g^{PRIM}, c_i, s)_k}_{\text{Consumption of ancillary energy in primary energy provision}} \cdot \underbrace{\frac{\sum_{c_i^*} \sum_{p_l^{CONV}} \sum_v \sum_s FLO_{in}(r, v, t, p_l^{CONV}, c_i^*, s)_k}{\sum_{c_i^*} \sum_{p_g^{PRIM}} \sum_v \sum_s FLO_{out}(r, v, t, p_g^{PRIM}, c_i^*, s)_k}}_{\text{Share of that primary energy carrier used for a certain conversion process}} + \underbrace{\sum_{p_l^{CONV}} \sum_v \sum_s FLO_{in}(r, v, t, p_l^{CONV}, c_i, s)_k}_{\text{Consumption of ancillary energy in final energy provision}} + \underbrace{\sum_{p_{c_i}^{DIST}} \sum_v \sum_s FLO_{in}(r, v, t, p_{c_i}^{DIST}, c_i, s)_k}_{\text{Consumption of ancillary energy in energy distribution}} \right] \cdot \underbrace{\gamma_k}_{\text{Share of the transport sector}} ; c_i \in FLO_{in}(p_g^{PRIM}, p_l^{CONV}, p_{c_i}^{DIST}) \wedge c_i^* \in pcg(p_g^{PRIM}) \quad (11)$$

(Table 3).  $c \in pcg(p)$  indicates commodities which are element of the primary commodity group of processes  $p$ . And with  $c_{GHG} = \{\text{CO}_2, \text{CH}_4, \text{N}_2\text{O}\}$ .  $DIST$  and  $CONV$  indicate a distribution or conversion process, respectively.  $PRIM$  indicates primary energy provision.  $k$  indicates the GHG taxes applied, with  $k = \{0, \dots, 250\}$  [EUR<sub>10</sub>/t CO<sub>2</sub>e].

Attributable to the activity of the transport sector is:

with

$\gamma_k$  = Transport share <sub>$k$</sub>

$$= \frac{TEP_{c_i}^{TRA}}{\sum_{c_i} \sum_p \sum_r \sum_v \sum_s FLO_{out}(r, v, t, p_l^{CONV}, c_i, s)_k} ; p \in p_l^{CONV} \wedge c_i \in pcg(p_l^{CONV})$$

**Table 3**  
Fuel provision options in the TIMES-GEECO model.

| Feedstock $c_{i*}$                           | Conversion process $p_i^{CONV}$                | Product $c_i$                    |
|----------------------------------------------|------------------------------------------------|----------------------------------|
| Crude oil                                    | Refining                                       | Petrol, diesel, LPG, kerosene    |
| Coal                                         | Gasification, Fischer Tropsch synthesis, (CCS) | Petrol, diesel                   |
|                                              | Gasification, upgrading, (CCS)                 | Substitute natural gas, hydrogen |
| Natural gas                                  | Compression                                    | CNG                              |
|                                              | Reforming, (CCS)                               | Hydrogen                         |
| Canola (rapeseed), soy beans, sunflower seed | Reforming, Fischer Tropsch synthesis, (CCS)    | Petrol, diesel                   |
| Waste cooking oil                            | Extraction, esterification                     | Biodiesel                        |
| Sugar Beet, Sugar, Cane                      | Preparation, esterification                    | Biodiesel                        |
| Landfill waste/Sewage sludge                 | Fermentation, distillation                     | Ethanol                          |
| Straw/wood                                   | Anaerobic digestion, upgrading                 | Substitute natural gas           |
|                                              | Gasification, Fischer Tropsch synthesis, (CCS) | Petrol, diesel                   |
|                                              | Gasification, upgrading, (CCS)                 | Substitute natural gas, hydrogen |
| Electricity                                  | Hydrolysis, fermentation, distillation         | Ethanol                          |
|                                              | (De-) centralized electrolysis                 | Hydrogen                         |

**Table 4**  
Overview of scenarios

| Scenario name                  | Scenario description                                                                                                                                                                               |
|--------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Business as usual (BAU)</b> | Business as usual politics. No GHG taxes applied                                                                                                                                                   |
| <b>Reference MAC</b>           | GHG taxes implemented from the year 2018 onwards                                                                                                                                                   |
| <b>Low energy price</b>        | Based on the assumption that global climate efforts can limit the growth of the crude oil price in the future, a scenario with lower primary energy carrier prices is analyzed (Table 5)           |
| <b>No CCS</b>                  | In this scenario, CCS technologies are assumed not to be available until 2040, to demonstrate the possible role of CCS for GHG mitigation in South Africa                                          |
| <b>Low battery price</b>       | In this scenario, specific battery investment costs have been reduced by 50% to 150 EUR <sub>07</sub> /kWh in 2040                                                                                 |
| <b>High biomass price</b>      | In this scenario the costs of providing lignocellulosic biomass (such as straw and wood) are increased by 50%. These energy carriers are important for the provision of second-generation biofuels |

Total transport related well-to-tank emissions can consequently be derived by multiplication of the attributable energy consumption with the corresponding emission factors as expressed in Eq. (12)

$$WTT_{GHGk}^{TRA} [gCO_2e] = \sum \left[ wtt_{c_i^*}^{PRIM} \cdot TEP_{c_i^*}^{TRA} \right] + \sum \left[ wtt_{c_i}^{CONV, DIST} \cdot \left[ TEP_{c_i}^{TRA} + TEC_{c_i}^{PRIM, CONV, DIST} \right] \right] \quad (12)$$

### 3. Scenario definition

To generate the various segments of the MAC curve (Eq. (1)) the TIMES-GEECO model is run with GHG taxes ( $k$ ) from 0 to 250 EUR<sub>10</sub>/t CO<sub>2</sub>e (in 18 model runs) implemented from the year 2018 onwards<sup>1</sup>. Smaller steps between the GHG taxes applied are chosen for lower abatement cost as most changes in the energy system are expected there. The model runs are then consolidated into a single MAC curve for the year 2040. GHG taxes have been increased over the model time-horizon (i.e., 2007–2040) with a discount rate of 8%, which is inherent to the model. All costs are given in 2010 values.

The calculations in this analysis exclude the application of taxes and subsidies (except of GHG taxes for generating the MAC curve) and indicate a macro-economic view; taxes reflect a re-allocation of resources and, thus, should not influence the optimal solution for a minimal-cost objective. When calculating the annuities, the technical lifetime of technologies is seen to be equal to the economic lifetime.

The future socio-economic development taken into account assumes a “mildly optimistic socio-economic scenario frame” for Gauteng (Wehnert et al., 2011). This vision adopts that in 2040 Gauteng will have become a megacity region with about 20 million inhabitants and will have experienced an approximately 3.7-fold increase in GDP. The income inequality in Gauteng in 2040 is still described as “dramatic”, but less than that in 2010 (Wehnert et al., 2011).

In order to show the robustness of the calculated MAC curve, several scenario variations are calculated (Table 4). The assumed development of primary energy carrier prices is based on the New Policies Scenario of the World Energy Outlook, which states an average crude import price of 123 USD<sub>10</sub>/bbl in the year 2040 (IEA, 2014). A low energy price scenario is calculated considering a crude oil price of 97 USD<sub>10</sub>/bbl in 2040. Assuming an on inflation of 2.3%/pa these figures translate into 243 and 192 USD<sub>40</sub>/bbl in 2040, respectively. All other primary energy carrier prices are aligned to the crude oil price based on historical price trajectories in South Africa available at DOE (2009) as shown in Table 5.

Carbon Capture and Storage (CCS) is considered as an option from 2020 onwards for all scenarios, according to CGS (2010) except for the “no CCS” scenario (Table 4). In this scenario, CCS

**Table 5**  
Prices for primary energy carriers for industrial consumers inclusive of transport and distribution.

|             | Unit                   | 2010 | 2040 | Low energy price scenario 2040 |
|-------------|------------------------|------|------|--------------------------------|
| Crude oil   | USD <sub>10</sub> /bbl | 78   | 123  | 97                             |
|             | EUR <sub>10</sub> /GJ  | 8.8  | 13.9 | 11.0                           |
| Coal        | EUR <sub>10</sub> /GJ  | 0.9  | 1.5  | 1.2                            |
| Natural gas | EUR <sub>10</sub> /GJ  | 7.2  | 11.9 | 9.5                            |

<sup>1</sup> All costs in this work are given in real terms for a base year of 2010.

technologies are assumed not to be available until 2040, to demonstrate the possible role of CCS for GHG mitigation in South Africa.

Lastly, the costs of some technologies and commodities, which are proposed as being promising measures for GHG mitigation, are varied in order to account for uncertainties. Thus, costs of providing lignocellulosic biomass (such as straw and wood) are increased by 50%, which is of reasonable magnitude given the uncertainty in provision costs found in the literature (see e.g., Damm and Triebel, 2008; Eisentraut, 2010 and DWAF, 2003). Additionally, specific battery costs specific battery investment costs have been reduced by 50% to 150 €/2007/kWh in 2040.

#### 4. Results and discussion

Under the assumptions of a business as usual development ( $k = 0$  EUR<sub>10</sub>/t CO<sub>2</sub>e) we see a steady growth in final energy consumption in the transport sector until 2040, which increases from 265 PJ in 2010 to about 510 PJ in 2040 (+92%). The dependence on fossil energy carriers is likely to continue. In the BAU scenario in 2040, the most important energy sources in the transport sector are crude oil-based products, namely petrol, diesel and kerosene which correspond to 80% (406 PJ) of its final energy consumption. Additionally, synthetic fuels provided via Fischer-Tropsch synthesis (FTS) of coal and natural gas are likely to still be a dominant energy carrier for transport in Gauteng in the future (i.e., 88 PJ in 2040). Biofuels have only minor significance in the BAU scenario in 2040 (2% of the transport final energy consumption) and originate mainly from biodiesel (source: waste cooking oil) and FTS fuels made out of biomass. Most important vehicle powertrains applied in the BAU scenario in 2040 are internal combustion engines, which account for 99.5% of the total vehicle activity in that year. However, those powertrains show clearly higher energy efficiency than today (inter alia due to start-stop systems and engine improvements applied), resulting in a decrease of specific final energy consumption by 38% between 2010 and 2040. Other vehicle powertrains remain negligible in the BAU scenario.

This continuation of using conventional fuel sources and powertrain technologies in conjunction with a significant increase in travel demand (i.e., +147% vkm in road transport) results in a

vast increase in transport well-to-wheel GHG emissions, which total at 51.5 Mt CO<sub>2</sub>e in 2040 (+58% in comparison to 2010).

Fig. 2 presents the result of the reference MAC curve for annual well-to-wheel GHG emissions of the transport sector. To better visualize the substitution effects due to the application of GHG taxes, the results have been shown as changes in comparison to the BAU scenario with a GHG tax of  $k = 0$  EUR<sub>10</sub>/t CO<sub>2</sub>e. The upper parts of the figures represent the emissions due to the application of alternative measures while the lower parts show how GHG emissions are mitigated.

At the lower end of the MAC curve only minor changes to vehicle powertrains are realized. These apply to hybrid electric engines used, for those modes showing longer annual driving distances such as buses, minibuses and heavy duty vehicles. Major changes, however, occur in transport fuel supply and use. Here, all Fischer-Tropsch synthetic fuels from coal to liquid (CTL) processes are replaced by other means for marginal abatement costs of 5–10 EUR<sub>10</sub>/t CO<sub>2</sub>e. Alternative fuels used are biodiesel made out of waste cooking oil and biomass-to-liquid (BTL) fuels. Additionally, electricity provision – which is mainly reliant on coal combustion under business as usual conditions – is decarbonized. Alternative power-generation technologies utilized at low marginal abatement costs are nuclear energy (3 EUR<sub>10</sub>/t CO<sub>2</sub>e), wind energy and coal CCS (5 EUR<sub>10</sub>/t CO<sub>2</sub>e) as well as concentrated solar power (10 EUR<sub>10</sub>/t CO<sub>2</sub>e). These measures result in a reduction of transport well-to-wheel GHG emissions of about 22.3 Mt CO<sub>2</sub>e at marginal abatement costs of 10 EUR<sub>10</sub>/t CO<sub>2</sub>e, corresponding to a decrease of 43% in comparison to the 2040 figure of 51.5 Mt CO<sub>2</sub>e under business as usual conditions.

The trend of electrification of the transport sector continues for more restrictive GHG mitigation policies. At moderate marginal GHG abatement costs of  $\leq 50$  EUR<sub>10</sub>/t CO<sub>2</sub>e, large shares of passenger car vehicle activity is attributed to mild hybrids. Additionally, full hybrid engines are applied to buses and heavy duty vehicles. To account for the increased contribution of ethanol from sugar cane (30 EUR<sub>10</sub>/t CO<sub>2</sub>e) and sugar beet (50 EUR<sub>10</sub>/t CO<sub>2</sub>e) also E85 flex-fuel engines are utilized. The application of these measures, however, only results in minor additional reduction of transport related GHG emissions. At marginal abatement cost of 50 EUR<sub>10</sub>/t CO<sub>2</sub>e, transport well-to-wheel emissions in 2040 amount to 25.5 Mt CO<sub>2</sub>e (–51% in comparison to the reference scenario).

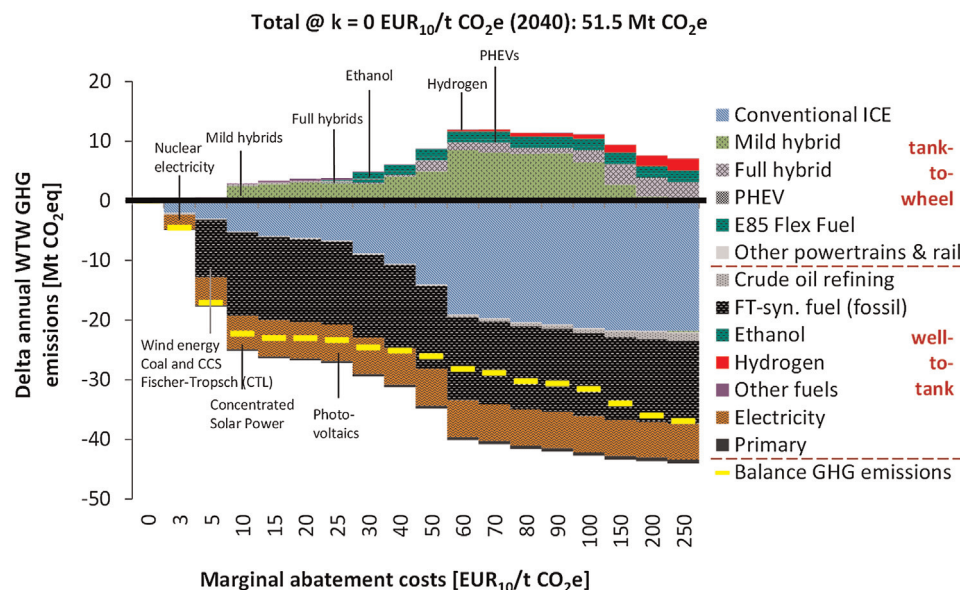


Fig. 2. Changes in annual well-to-wheel GHG emissions attributable to the Gauteng transport sector compared to a business as usual case without GHG taxes in 2040. "Other powertrains" comprise biodiesel, natural gas and hydrogen combustion engines. "Other fuels" comprise biodiesel, biomass-to-liquid and methane gas.

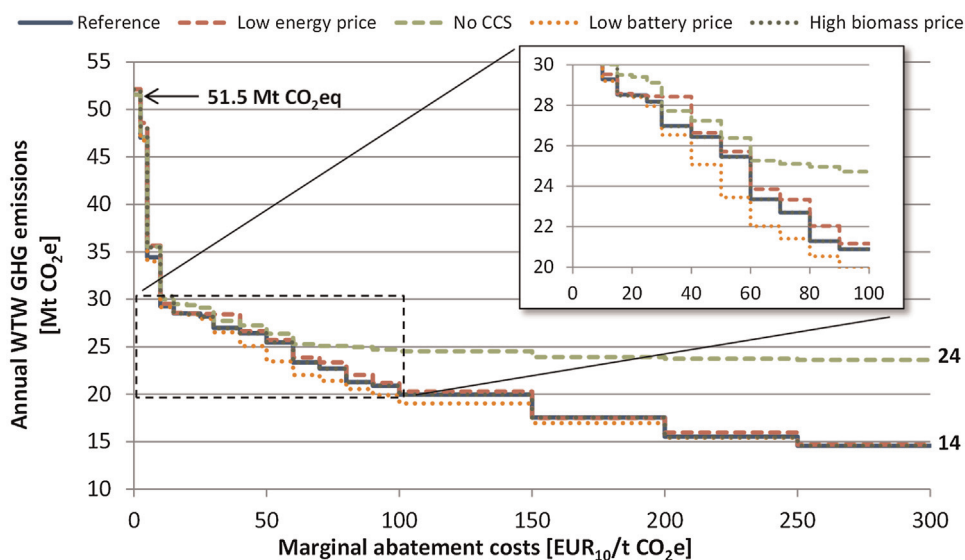


Fig. 3. Marginal abatement costs of transport related GHG emissions in Gauteng (well-to-wheel) in 2040 for different scenarios.

At the upper end of the calculated MAC curve ( $> 50$  EUR<sub>10</sub>/t CO<sub>2</sub>e), remaining conventional powertrains are replaced. On the one hand, electrification is intensified with applying full hybrid vehicles for passenger cars (100 EUR<sub>10</sub>/t CO<sub>2</sub>e), light duty vehicles (200 EUR<sub>10</sub>/t CO<sub>2</sub>e) and SUVs (250 EUR<sub>10</sub>/t CO<sub>2</sub>e) as well as plug-in hybrid electric vehicles for minibuses and light duty vehicles ( $> 70$  EUR<sub>10</sub>/t CO<sub>2</sub>e). The application of PHEVs also results in increased consumption of electricity in the transport sector, which shows a low emission factor (i.e. about 40 g CO<sub>2</sub>e/kWh) at these marginal abatement cost. Additionally, fuel-cell hybrid electric vehicles become a more and more central powertrain concept at increasing marginal abatement cost. Hydrogen is initially supplied via natural gas reforming at marginal abatement costs of 60 EUR<sub>10</sub>/t CO<sub>2</sub>e and at 250 EUR<sub>10</sub>/t CO<sub>2</sub>e also via biomass gasification. At marginal abatement cost of 250 EUR<sub>10</sub>/t CO<sub>2</sub>e, transport well-to-wheel GHG emissions can thereby be reduced to 14.6 Mt CO<sub>2</sub>e, which is 72% less than under business as usual conditions in 2040.

The sensitivity analysis shows that most of the options identified for GHG mitigation are not sensitive, except those technologies using carbon capture and storage, such as hydrogen from coal gasification and upgrading. Ignoring carbon capture and storage technology discourages the uptake of hydrogen and limits the use of electricity in the transport sector, as the provision of low-carbon electricity is more expensive. Total well-to-wheel emissions are greater, especially at higher marginal abatement cost when not using carbon capture and storage (Fig. 3).

Lower crude oil prices are responsible for comparably higher annual well-to-wheel GHG emissions for most marginal abatement costs as they impede the substitution of conventional fuels and powertrains, although this effect diminishes for higher abatement costs. On average, the well-to-wheel GHG emissions are about 0.5 Mt CO<sub>2</sub>e higher than in the reference MAC curve.

As the costs of biomass provision increase, the uptake of BTL is slowed down, but it can still be seen as a feasible option for GHG mitigation. Cheaper batteries encourage hybrid electric vehicles. However, the differences in GHG emission mitigation are of little account.

## 5. Conclusion and policy implications

The transport sector globally accounts for significant greenhouse gas emissions. Various options are discussed in order to

reduce further emission growth or to even mitigate climate relevant gases to a more sustainable level. Natural resources, such as fossil fuel sources but also available biomass, are limited as well as the financial opportunities of a country. Thus, the question arises which measures should be taken to not exceed acceptable GHG abatement costs. Marginal abatement cost curves are one tool to relate emission mitigation levels with costs and allow identifying the corresponding set of measures. Several options exist in order to create marginal abatement costs curves, but energy system analysis allows including the interlinkages between measures and interring temporal aspects and thus providing a consistent view.

To overcome some limitations of existing approaches a new method was presented, aiming to make model results more transparent. Making use of the TIMES-GECCO model, which is an application of the energy system model generator, TIMES, for Gauteng Province and South Africa, this study shows how to visualize the changes and substitution effects in marginal abatement cost curves. By attributing sectoral effects in energy provision to energy use to the transport sector it is furthermore demonstrated how to derive marginal abatement costs curves for well-to-wheel emissions for the transport sector considering the full energy provision chain and the interlinkages and interdependencies within the energy system. The most relevant greenhouse gases were included in this analysis.

This analysis has shown that, under current political dispensations, the GHG emissions and energy consumption of Gauteng are likely to increase significantly. Policymakers should therefore intervene at an early stage to influence or dictate desirable developments. For the example of Gauteng province, this study showed that the transport sector is not the first sector to be addressed for cost-efficient reduction of GHG emissions. However, the analysis also showed that several options are available to mitigate transport related GHG emissions at comparable low marginal abatement costs.

It has been concluded that initial changes in the energy system should involve the provision of electricity by switching from conventional coal-fired power plants to alternatives such as nuclear, solar energy (e.g., large-scale concentrated solar power collectors using parabolic troughs, but also roof-mounted photovoltaics units on buildings), or coal in conjunction with carbon capture and storage. However, currently less than 5 PJ of electricity are used in the transport sector of Gauteng (i.e., for rail transport) which impedes a mitigation potential below 2 Mt CO<sub>2</sub>e. Further

electrification of the transport sector would obviously increase the potential for GHG mitigation.

The substitution of fossil Fischer-Tropsch synthetic fuels is relatively easily achievable as even crude oil-based fuels are responsible for significantly lower well-to-wheel emissions. However, further issues, such as those of supply security, should be taken into account as coal to liquid fuels currently provide a significant share of transport fuels.

On the other hand, the opportunities for Gauteng's government to initiate such changes might be limited. One option is to tender for the provision of "green" electricity, which might be imported to Gauteng. National government could intervene directly with measures like GHG taxes for fuels and electricity. At least from a GHG mitigation point of view, no additional coal to liquid capacity should be built without using carbon capture and storage and no new coal power plants without being CCS-ready.

In the Gauteng transport sector, local government can actively reduce the province's climate impact by encouraging the use of hybrid electric vehicles, especially for those modes which cover substantial distances, i.e. buses. As most of the bus services are state owned, policy-makers can directly affect the adoption of hybrid vehicles for that mode. Moreover, recapitalization programs offer the possibility of subsidizing hybrid vehicles. As minibuses transport is the backbone of the public transport system of the province the government should address this fact, by integrating the minibus sector into the transport system instead of discouraging minibus transport. An option might be to regulate the minibus sector and its integration into a combined public transport system.

The effect of fleet renewal on specific energy consumption in the transport sector can be significant. Policymakers can accelerate the introduction of vehicles with lower fuel consumption and better emission levels by stricter emission legislation and scrapping schemes. National support can help to implement such policies, for example, with minimum requirements for imported or new registered vehicles. Once again, local government could play a guiding role by using vehicles with better standards for their own fleet and for public vehicles in the province.

Furthermore, certain biofuels offer the potential of "greening" the transport sector at comparatively low costs. Initial measures could include the use of biodiesel from waste cooking oil and ethanol from sugar cane. Biodiesel from waste cooking oil can be produced within the province. The government could support its adoption by preventing the waste cooking oil being used as animal food and giving tax reductions for such biofuels. Furthermore, research that considers the quality of the fuel produced should be supported. Additionally, the provincial government can fund the upgrading of gas from landfill sites and sewage plants as incentives may be required to support the use of such energy carriers on a commercial scale. However, to achieve significant reduction potentials, the gas of most big plants would have to be used, which could be achieved by appropriate legislation. Landfill and sewage gas for electricity provision competes with its upgrading to a substitute for natural gas. The choice of either option is obviously dependent on many factors, such as the economies of scale, the distance of the site to the grid or distribution station, the biogenic share in the waste and the cost of gas upgrading. Moreover, taxes and the reference price of electricity and fuels apparently affect the optimal choice for commercial producers. In this study, taxes were not considered as part of a macro-economic view.

Significant GHG reduction potential can be seen in second-generation biofuels, which might be produced outside the boundaries of Gauteng. In the scenario analysis, biomass-to-liquid was the preferred option. On the other hand, ethanol from lignocellulosic biomass showed comparably low well-to-wheel GHG

emissions and fuel costs in the long term, which indicates that this option should not be neglected. However, the necessary production technologies for second-generation biofuels are currently not commercially available and so their future feasibility is uncertain. The provincial and national governments should therefore encourage research in all fields of second-generation biofuel production and sponsor demonstration projects.

These comprise of biofuels made out of waste cooking oil and cellulosic biomass. Characteristic for the energy system of South Africa is the provision of synthetic fuels from fossil sources, whose substitution would significantly mitigate GHG emissions. Alternative powertrains, such as hybrid electric vehicles, applied for transport modes showing comparably long annual driving distances (e.g., buses), also shows low GHG abatement costs, but in the case of Gauteng the mitigation potential is low.

This study involved a comprehensive analysis of possible developments and potential alternative pathways for GHG reduction for the transport sector and the first application of a cost-optimizing energy system model for Gauteng. The presented method can also be applied to other socio-economic sectors or the whole energy system to give a more comprehensive view of possible GHG mitigation options for Gauteng and South Africa. The method presented, could furthermore be transferred to other regions in the world to visualize the dependencies and interlinkages between GHG mitigation measures allowing to derive cost-efficient greenhouse gas reduction strategy.

Although this study indicates quantitative figures for GHG abatement costs, it is made clear that these are an image for the purpose of taking action in one specific point in time and depending on country specific characteristics of the energy system (e.g., the use of fuels based on the coal to liquid process in South Africa). Moreover, derived abatement costs can depend on several factors such as the model type applied, the sectoral and technology details included in the analysis, the assumptions taken for energy carrier prices, consideration of technology learning or the inclusion of greenhouse gases besides carbon dioxide. Notably, scientific literature is often contradicting and finds arguments to support and to disprove the influence of these factors. Other assumptions such as the discount rate taken for the analysis as well as the exclusion of certain mitigation options obviously influence the marginal abatement costs (see e.g., [Kesicki, 2013](#); [Klepper and Peterson, 2006](#); [Flachsland et al., 2011](#)).

Future analysis might also include a more detailed view of non-GHG emissions by including externalities and social costs, which might prove to be important, especially in an urban environment like Gauteng. The industry sector of the province has been identified as a major consumer of energy that causes a significant impact on climate. Future research is needed to identify least-cost options in this sector. For example, similar to the transport sector, fossil fuel based Fischer-Tropsch synthesis in basic chemical production could partially be replaced by alternative processes, which would be an enhancement of current energy system models. In the residential sector, the characterization of the population by income group will allow the analysis of the effects of economic transformation and enable the finding of applicable and feasible measures to mitigate GHG emissions in that sector. Public buildings have so far not been analyzed in detail in the TIMES-GECCO model. Even though the public sector accounts for only a small (< 1%) fraction of total energy consumption and GHG emissions, government buildings can become a role model to demonstrate the implementation of GHG mitigation measures.

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